

CUSTOM EXHIBITION NAVIGATION & LIVE TRACKING SYSTEM

Comprehensive Project Plan — Version 1.0 (Full Detailed Specification)

Project Goal

Develop a web application that allows visitors to navigate through exhibitions, stores, campuses, or private areas using custom paths instead of relying on Google Maps.

The system should provide:

- Live GPS tracking
- Custom navigation paths
- Store searching
- Exhibition information
- Live announcements
- Voice broadcasting (future)
- Admin management system
- Real-time updates

Technology Stack

Frontend

- React.js (Vite)
- HTML5
- CSS / Tailwind CSS
- JavaScript / TypeScript
- Leaflet.js
- OpenStreetMap

Backend

- Supabase
- PostgreSQL
- Supabase Realtime
- Supabase Authentication
- Supabase Storage
- Edge Functions (if required)

Hosting & Repository

- **Frontend Hosting:** Cloudflare Pages
- **Backend Hosting:** Supabase Cloud
- **Repository:** GitHub
- **Version Control:** Git + GitHub

Project Modules

1. User Module

- Login
- Register
- Anonymous visitor mode
- User profile
- GPS permission
- Location updates

5. Navigation Module

- Detect user location
- Find nearest node
- Calculate shortest path
- Draw navigation line
- Live position updates
- Destination guidance

2. Admin Module

- Dashboard
- Manage exhibitions
- Manage stores
- Manage categories
- Manage navigation nodes
- Manage routes
- View visitor statistics
- Send announcements

3. Store Module

- Store information
- Images
- Promotions
- Categories
- Opening hours
- Contact details
- Search stores

4. Exhibition Module

- Exhibition details
- Event schedule
- Map
- Categories
- Featured exhibitions

6. Search Module

- Search stores
- Search exhibitions
- Search facilities
- Search services

7. Notification Module

- Push notifications
- Live announcements
- Emergency alerts

8. Analytics Module

- Visitor count
- Popular stores
- Most visited routes
- Heat map
- Live visitor statistics

Database Design (Tables & Fields)

Table Name	Fields
Users	id, name, email, phone, role, created_at
Exhibitions	id, title, description, image, start_date, end_date
Stores	id, name, description, logo, category_id, latitude, longitude, opening_time, closing_time
Categories	id, name
Navigation_Nodes	id, name, latitude, longitude
Navigation_Edges	id, from_node, to_node, distance

Table Name	Fields
Routes	id, store_id, node_id
Announcements	id, title, message, created_at
Visitor_Locations	user_id, latitude, longitude, updated_at

Frontend Pages	Frontend Components	Backend APIs
• Home Page • Login Page • Register Page • Map Page • Navigation Page • Store Details • Exhibition Details • Search Page • Profile Page • Admin Dashboard • Settings • About	• Navbar • Sidebar • Map • GPS Button • Store Card • Exhibition Card • Search Bar • Navigation Panel • Bottom Navigation • Profile Card • Notification Popup • Loading Screen	• Authentication • Store APIs • Exhibition APIs • Navigation APIs • Search APIs • Location APIs • Realtime APIs • Announcement APIs • Admin APIs

Navigation Algorithm	Map Features	Future Features
1. Get current GPS location 2. Find nearest navigation node 3. Calculate shortest path Algorithm: ◦ Dijkstra OR A* 4. Return route 5. Draw route on map 6. Update user location continuously	• Current location • Destination marker • Store markers • Exhibition markers • Facilities • Parking • Washrooms • Food courts • Emergency exits • Zoom controls • Compass • Navigation line	• Indoor navigation • QR Code checkpoints • Bluetooth Beacon positioning • Offline maps • Voice navigation • AI route recommendations • Crowd density prediction • Traffic estimation • Store recommendations • Multi-language support • Dark Mode • Accessibility Mode

Voice Broadcasting (Future Implementation)

Broadcaster → Icecast Server → All visitors receive live announcements

Examples: "Hall A presentation starts in 5 minutes." | "Emergency exit is on the left."

Security

- HTTPS
- Supabase Authentication
- Row Level Security
- Admin permissions
- Input validation
- Rate limiting

Testing

- GPS accuracy testing
- Navigation testing
- Load testing
- Database testing
- Mobile testing
- Cross-browser testing

Deployment & Mobile App

GitHub → Cloudflare Pages → Supabase → Production

Possible Future Mobile App:
Flutter or React Native using the same Supabase backend.

Project Timeline

- **Phase 1:** Project setup, GitHub, Cloudflare, Supabase
- **Phase 2:** Authentication, Database, User management
- **Phase 3:** Map integration, GPS tracking, Current location
- **Phase 4:** Custom navigation system, Nodes, Edges, Shortest path
- **Phase 5:** Store management, Exhibitions, Search
- **Phase 6:** Realtime location updates
- **Phase 7:** Announcements, Notifications
- **Phase 8:** Testing
- **Phase 9:** Deployment
- **Phase 10:** Future improvements

Recommended Repository Architecture & Scalability

```
project-root/
|
├── frontend/           # React + Vite
├── backend/            # Edge Functions (if needed)
├── database/           # SQL schema and migrations
├── docs/               # Documentation
├── assets/             # Images, icons
├── scripts/            # Utility scripts
├── tests/              # Automated tests
└── README.md
```

Additional Future Modules for Production

- QR code navigation (scan a QR code to instantly locate the user indoors)
- Volunteer/staff tracking
- Event scheduling and reminders
- Emergency evacuation routing
- Heat maps showing crowded areas
- Business analytics for exhibitors

- AI-powered recommendations (e.g., “You may also like these nearby stores”)
- Offline support using Progressive Web App (PWA) features.